

Characteristics of polynomial function



- 1 a -8 b -1 c 8184 d $16\,376$
- 2 a $\frac{17}{6}$ b $-\frac{16}{3}$ c $\frac{1529}{384}$ d 4
- 3 a 20 b $\frac{131\,950}{6561}$ c -1932 d 42
- 4 a Fall b Fall
- 5 It is impossible. A polynomial function that has degree n has at most $n - 1$ turning points.
- 6 A polynomial function that has degree n has a graph with at most $n - 1$ turning points.
- 7 a $(0, 5)$ b 2
- 8 a Yes b No
- 9 False
- 10 a No b No c No d Yes



Graphs of polynomial functions

11

a

i $x = 3, -1$

iii $(-1, 3)$

ii $(-\infty, -1), (3, \infty)$

b

i $x = -3, -1, 1$

iii $(-3, -1), (1, \infty)$

ii $(-\infty, -3), (-1, 1)$

c

i $x = -4, -2, 2$

iii $(-\infty, -4), (-2, 2)$

ii $(-4, -2), (2, \infty)$

d

i $x = 0, 2$

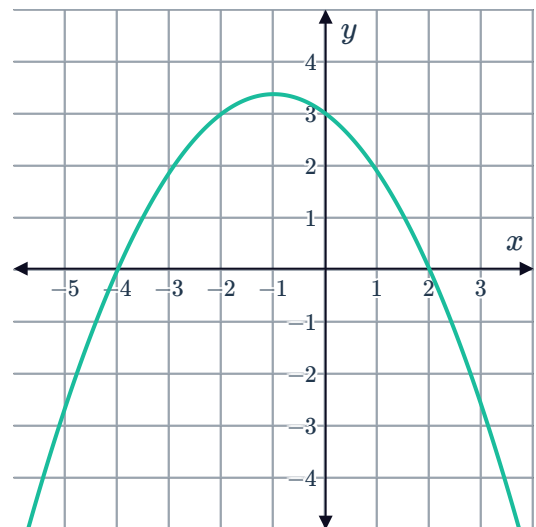
iii $(0, 2)$

ii $(-\infty, 0), (2, \infty)$

12

a 2

b



13

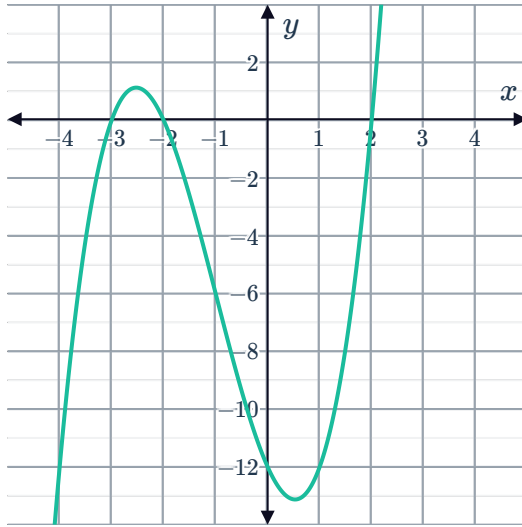


a

i $x = -3, -2, 2$

ii $y = -12$

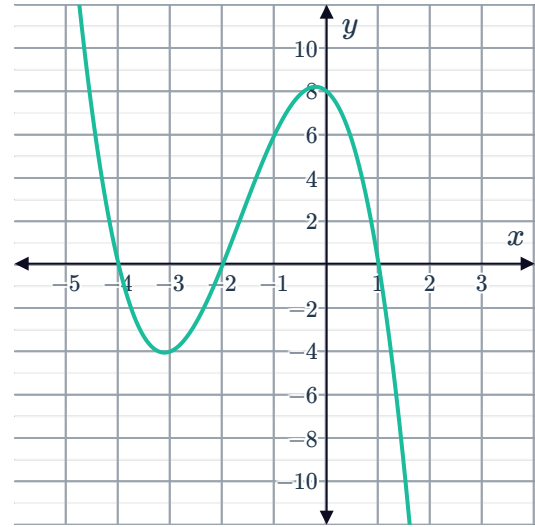
iii



i $x = -4, -2, 1$

ii $y = 8$

iii

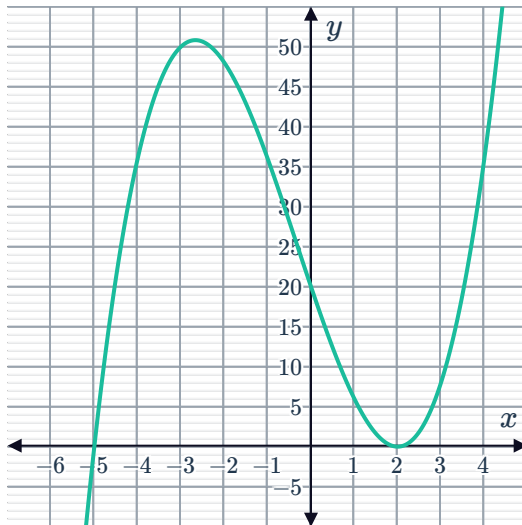


c

i $x = 2, -5$

ii $y = 20$

iii

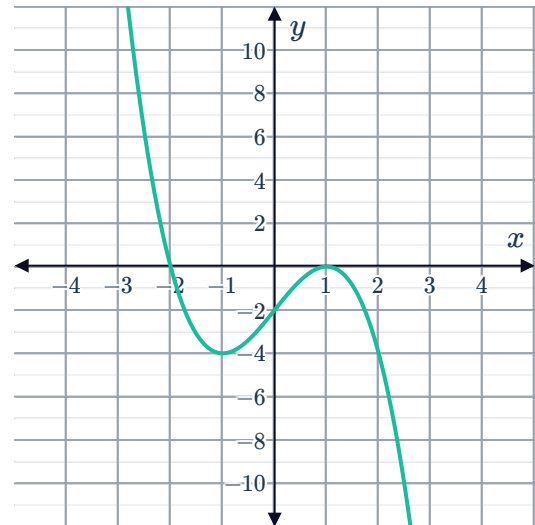


d

i $x = 1, -2$

ii $y = -2$

iii

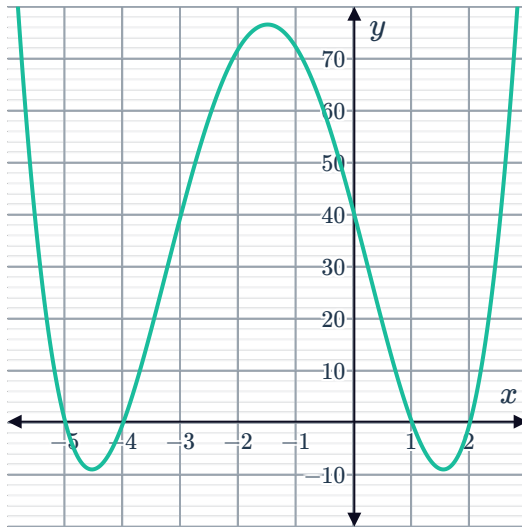


e

i $x = 1, 2, -4, -5$

ii $y = 40$

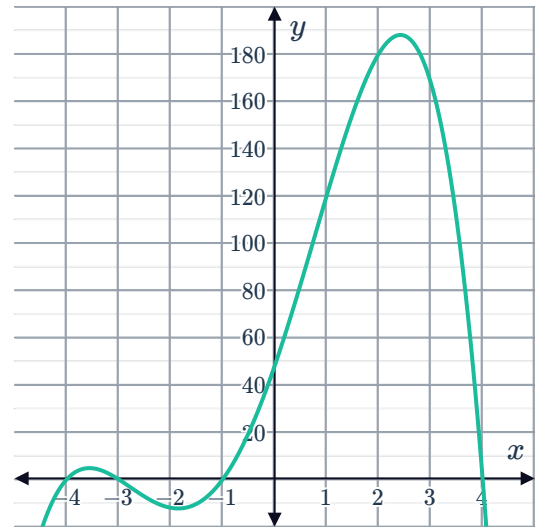
iii



i $x = -1, -3, -4, 4$

ii $y = 48$

iii

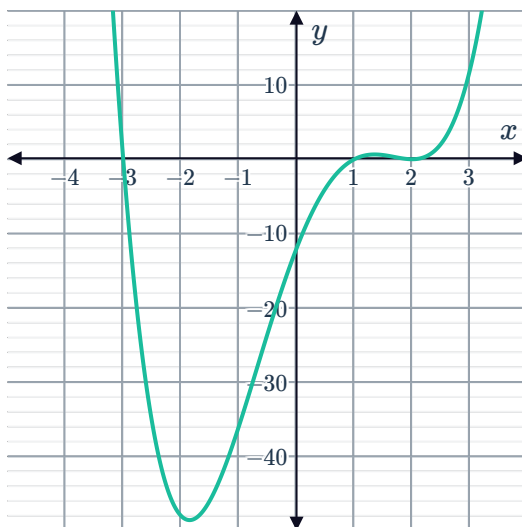


g

i $x = 2, -3, 1$

ii $y = -12$

iii

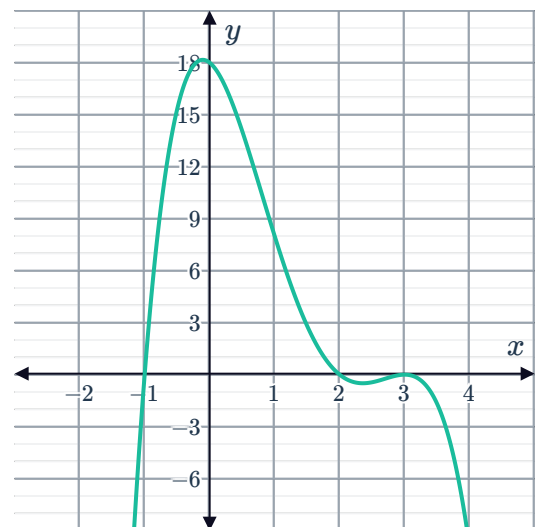


h

i $x = 3, -1, 2$

ii $y = 18$

iii

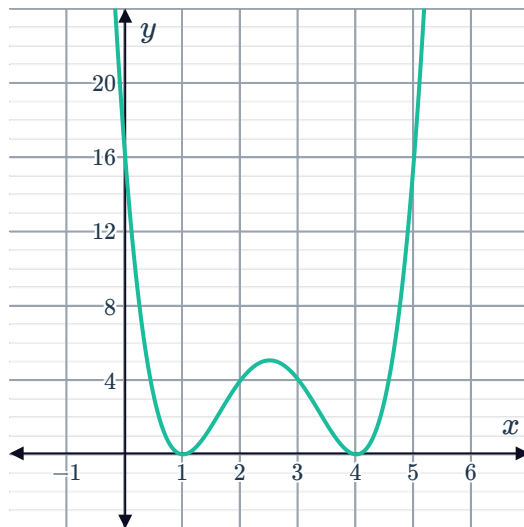


i

i $x = 1, 4$

ii $y = 16$

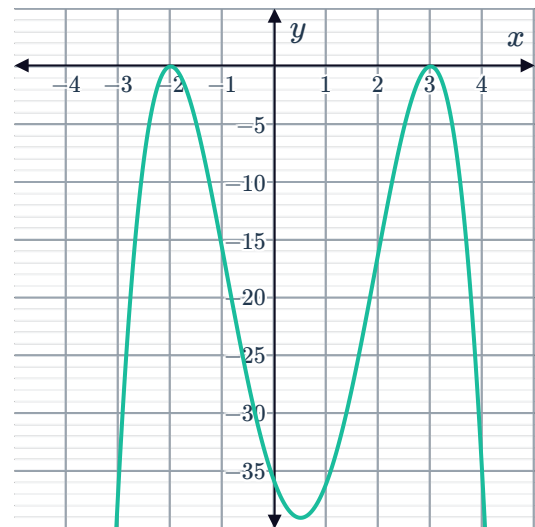
iii



i $x = 3, -2$

ii $y = -36$

iii



14

a Rise

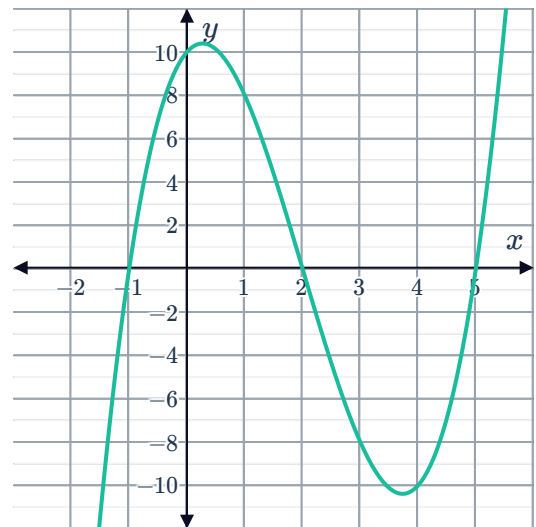
c $y = (x - 2)(x - 5)(x + 1)$

e $y = 10$

b Fall

d $x = 2, 5, -1$

f



15

a 1

c Rise

e $x = 0, -2, 2$

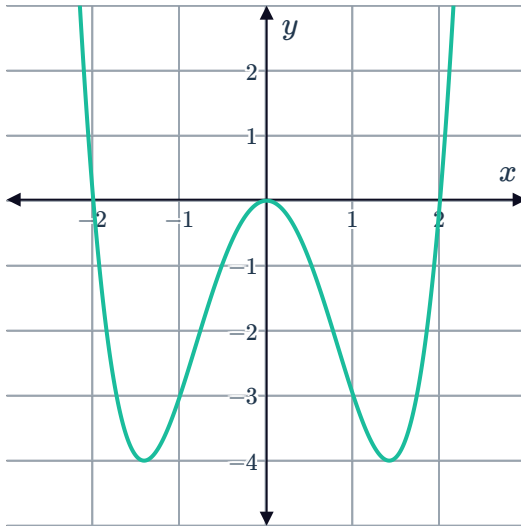
g



b Rise

d $y = x^2(x - 2)(x + 2)$

f $y = 0$



16

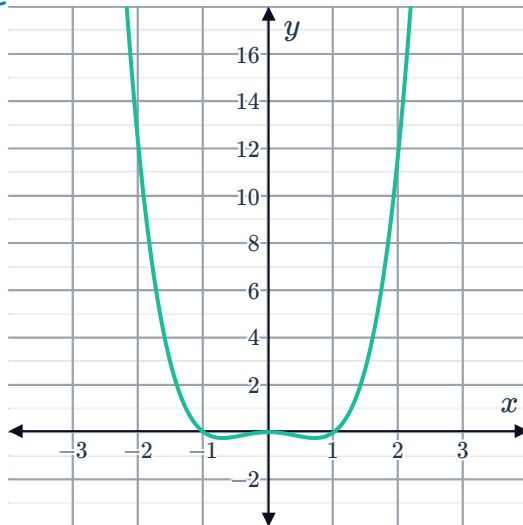
a 1

b Even

c Rise

d Rise

e





a

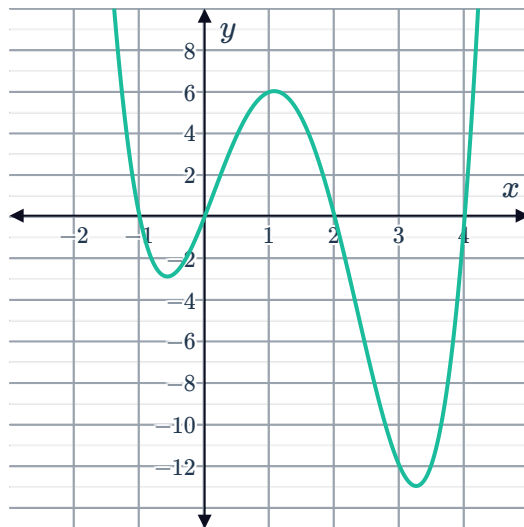
i $y \rightarrow \infty$

ii $y \rightarrow \infty$

iii $x = 0, -1, 2, 4$

iv $y = 0$

v



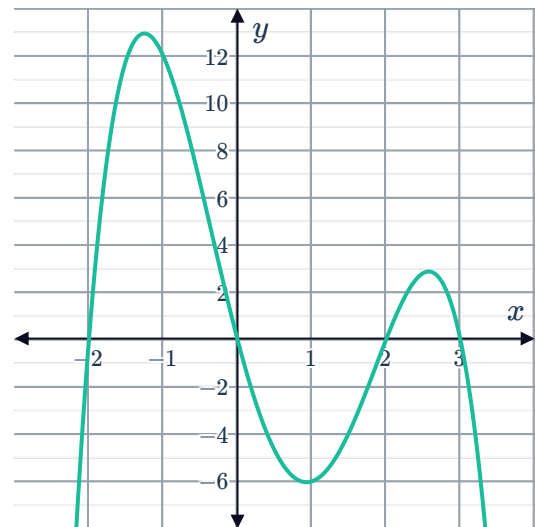
i $y \rightarrow -\infty$

ii $y \rightarrow -\infty$

iii $x = 0, -2, 2, 3$

iv $y = 0$

v



c

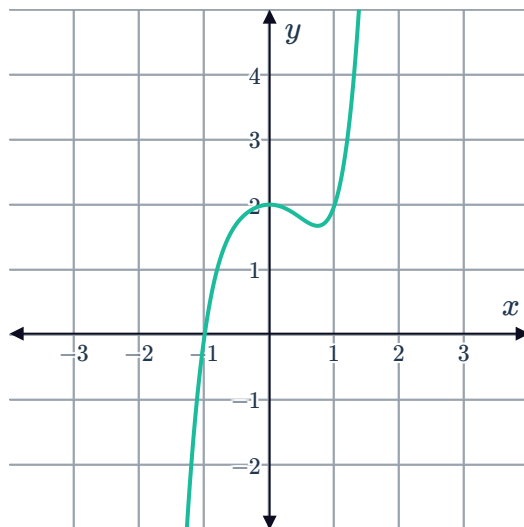
i $y \rightarrow -\infty$

ii $y \rightarrow \infty$

iii $x = -1$

iv $y = 2$

v



d

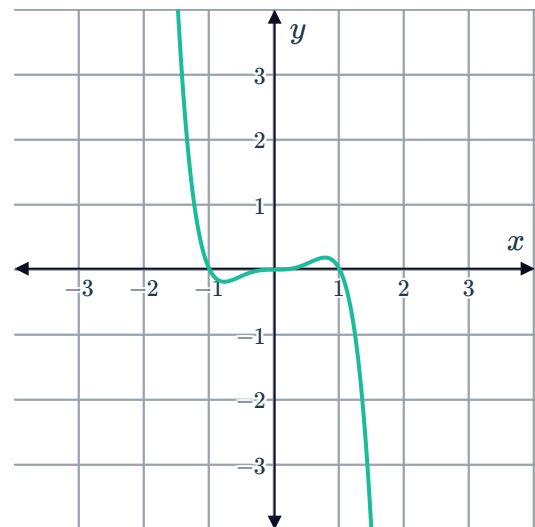
i $y \rightarrow \infty$

ii $y \rightarrow -\infty$

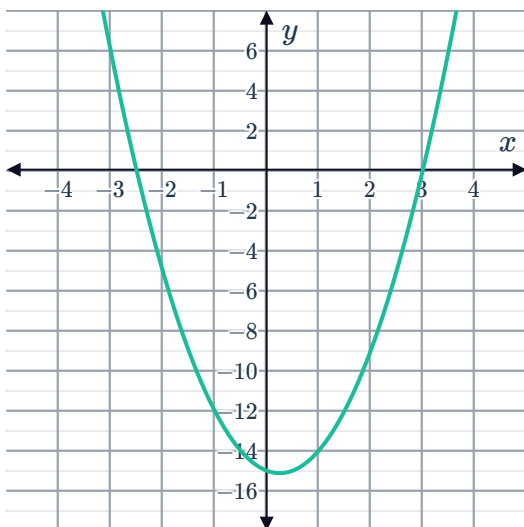
iii $x = 0, 1, -1$

iv $y = 0$

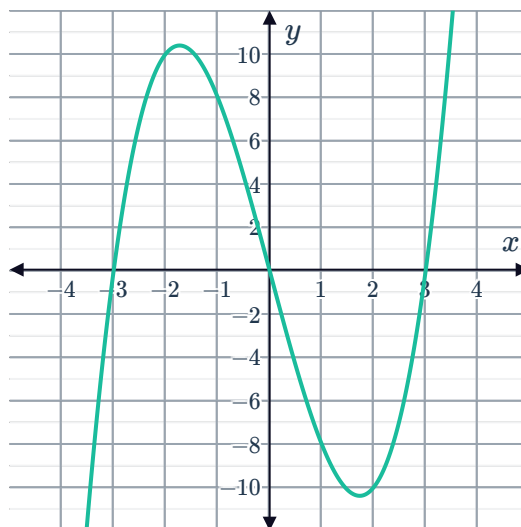
v



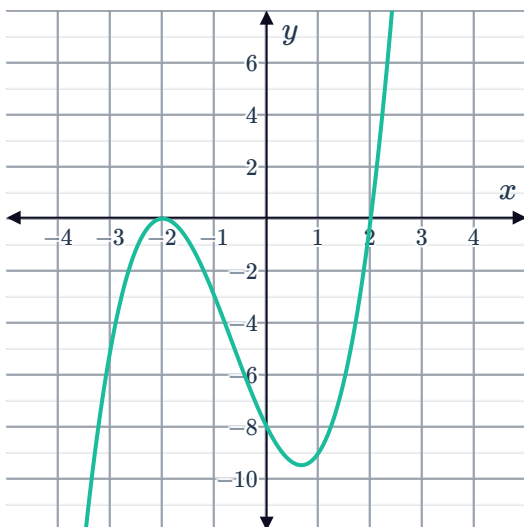
a



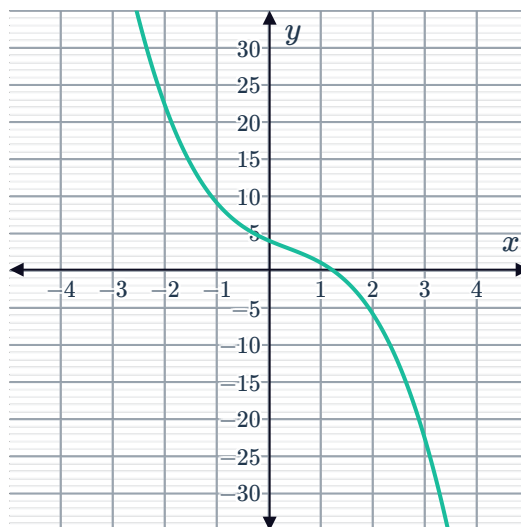
b



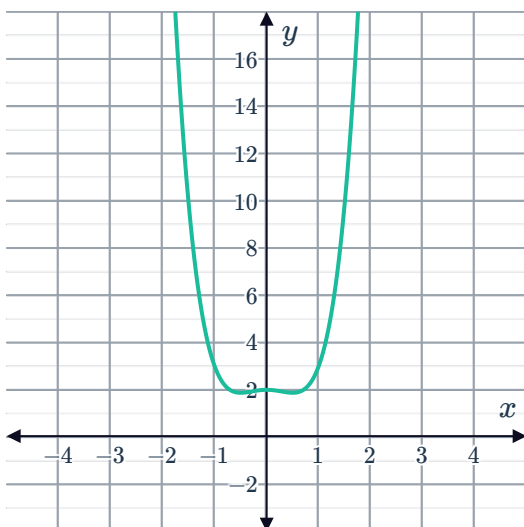
c



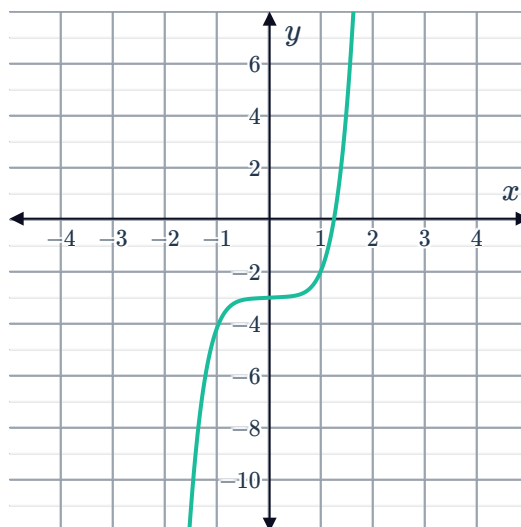
d



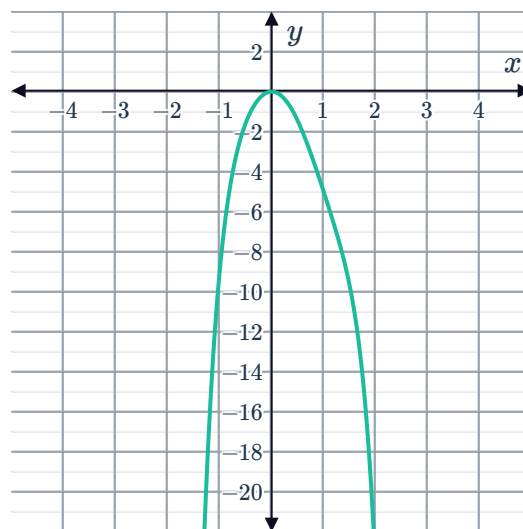
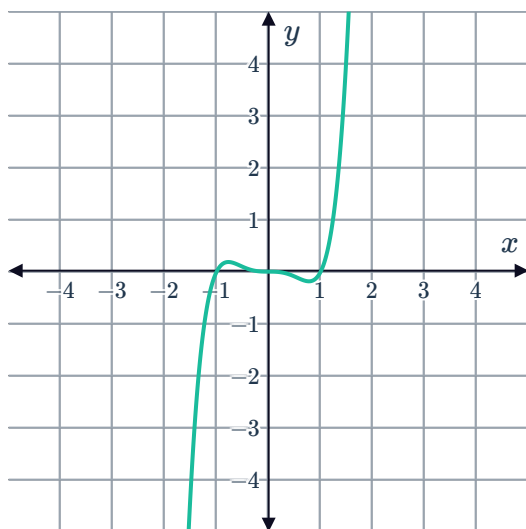
e



f



g



Applications

19

- a 54.24. At 23 years of age, a woman's level of good cholesterol should be around 54.24 mg/dL.
- b 57.88. At 51 years of age, a woman's level of good cholesterol should be around 57.88 mg/dL.

20

- a 16.658 L b 22.718 L c 25 years d 1995

21

- a 9420 b 1081 c 56 years d 1986

22

- a 390 m b 0 m

23

- a 180 ft b 7 seconds

24

- a $P = x^3 + 2x^2 - 10x + 7$ b Profit

25



a

r	1	2	4	5	7
V	$\frac{4\pi}{3}$	$\frac{32\pi}{3}$	$\frac{128\pi}{3}$	$\frac{500\pi}{3}$	$\frac{1372\pi}{3}$

b

- i No ii Yes iii No iv No

c $r = 6$ m

26

a 8 kg

b 44.48 cm

c 21 containers

27

a $V = 4x(3 - x)(5 - x)$ b $0 < x < 3$

c

d 24 ft^3 